



Roll No. ...5019932....

97763

BCA 4 Years 3rd Semester (NEP)
Examination – December, 2025

OBJECT-ORIENTED PROGRAMMING USING C

Paper : 24BCA403DS02

Time : Three Hours]

[Maximum Marks : 50

39

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *five* questions in all, selecting *one* question from each Unit. Question No. 1 is *compulsory*. All questions carry equal marks.

1. Explain the following in short :

5 × 2 = 10

- (a) Encapsulation
- (b) Destructor
- (c) Static data members



(d) Pure virtual functions

(e) File modes

UNIT – I

2. In what ways is an object-oriented paradigm better than a structured programming paradigm ? Explain the features of OOPs.

10

3. (a) How a class is different from structure ?

2

(b) Discuss various types of loops in C++ with examples.

8

UNIT – II

4. What do you mean by Inheritance ? Describe single and hybrid inheritance with suitable examples.

10

5. What do you mean by constructor ? Explain types of constructors with examples.

10



UNIT – III

6. What do you mean by operator overloading ? What are various restrictions in operator overloading ? Write a program in C++ to overload any unary operator. 10
7. (a) What do you mean by virtual function ? What is their importance ? 4
- (b) What do you mean by polymorphism ? Explain with the help of example how polymorphism is achieved ? 6

UNIT – IV

8. (a) Discuss try, throw and catch with reference to exception handling. 5
- (b) What is an exception ? Give some scenarios that cause exceptions. 5

9. Write notes on :

(a) Various file handling operations

5

(b) Generic programming

5

